

DRDO PASSIVE SHELTER THERMAL DESIGN REPORT

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1. Executive Summary & Location Climate

Shelter Name:	High-Altitude Passive Shelter
Location / Climate Zone:	Lahaul, Lahaul and Spiti, Himachal Pradesh, India
Location Ambient Bounds:	-16.9°C (Min) to 6.9°C (Max)
Predicted Optimal Shelter Shape:	Dome

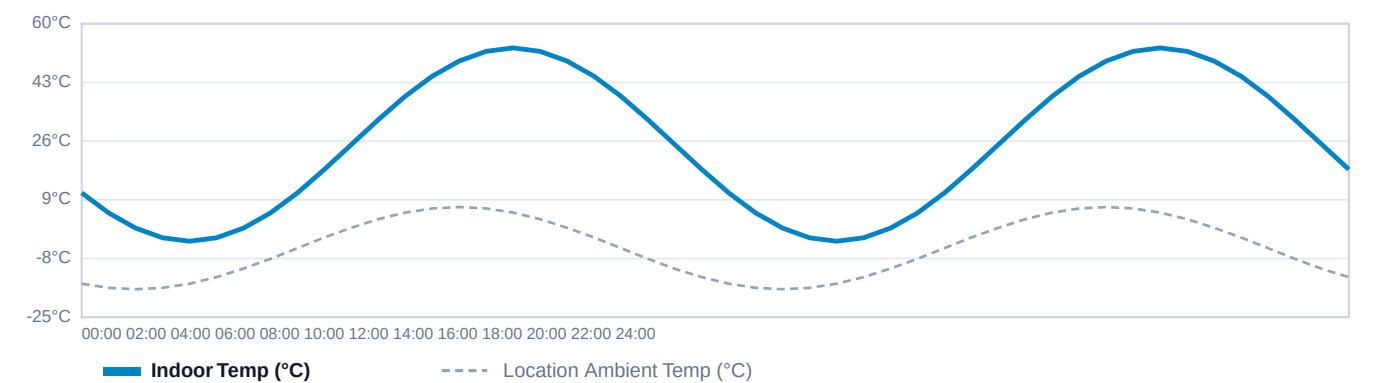
2. Shelter Model Overview & Envelope Materials

Envelope Component	Selected Material	Conductivity (k)	Thickness
Wall Envelope	Straw-Clay Composite	0.22 W/mK	0.25 m
Roof Structure	Sandwich PUF Roof Panel	0.024 W/mK	0.2 m
Floor Slab	Insulated Concrete Floor Slab	1.1 W/mK	0.15 m
Thermal Insulation	Polyurethane Foam (PUF) Insulation	0.024 W/mK	0.10 m
Glazing Windows	Double Low-E Argon Glazing	1.2 W/mK	2 units (2.5m²)
Entrance Doors	Insulated Solid Timber Door	0.13 W/mK	1 units (1.8m²)

3. Thermal Performance KPI Metrics

Performance Metric	Calculated Value	Engineering Target / Units
Average Indoor Temperature	50.7 °C	Target: > 15 °C
Minimum Indoor Temperature	11.7 °C	Sub-zero minimum
Maximum Indoor Temperature	60 °C	Solar midday peak
Thermal Comfort Index (18-24°C)	1.2 %	Percent of hours in comfort zone
Total Solar Energy Captured	343.45 kWh	Cumulative passive solar gain
Total Conduction & Sky Loss	251.56 kWh	Cumulative thermal loss
Peak Thermal Heat Loss	1799.9 W	Maximum heating load

4. 24-Hour Indoor vs. Ambient Temperature Profile



5. Envelope Heat Loss Component Breakdown

Envelope Component	Average Conduction Loss (W)	Share of Total Loss (%)
Walls Conduction	391.3 W	30.1 %
Roof Conduction	88.5 W	6.8 %
Floor Conduction	296.8 W	22.9 %
Windows Conduction	378.1 W	29.1 %
Doors Conduction	143.7 W	11.1 %

6. Engineering Recommendations

The predicted Dome shelter design optimizes thermal resistance and solar gain for the specified climatic location. Additional PUF insulation panels are recommended for sub-zero night hours.